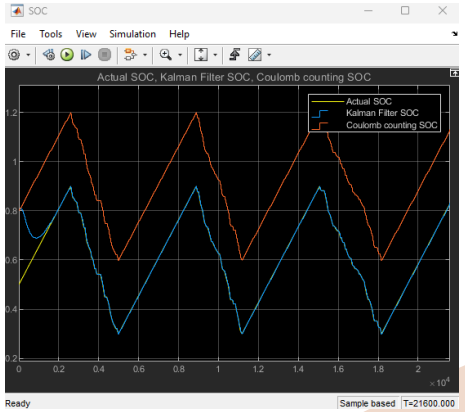
Simulating an Electric BAJA Vehicle with MATLAB and Simulink

Veer Alakshendra, MathWorks

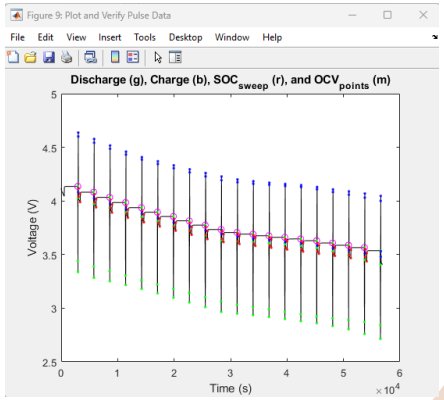
© 2025 The MathWorks, Inc.

What will you learn today?

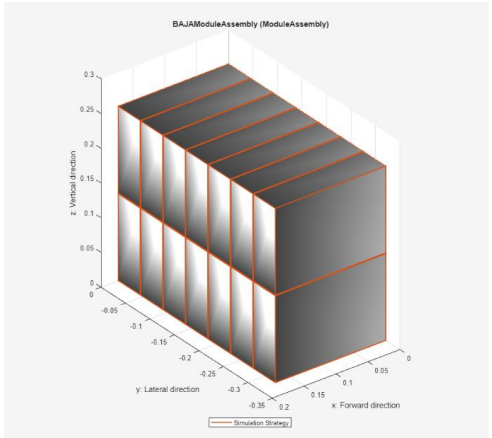


Electric powertrain design

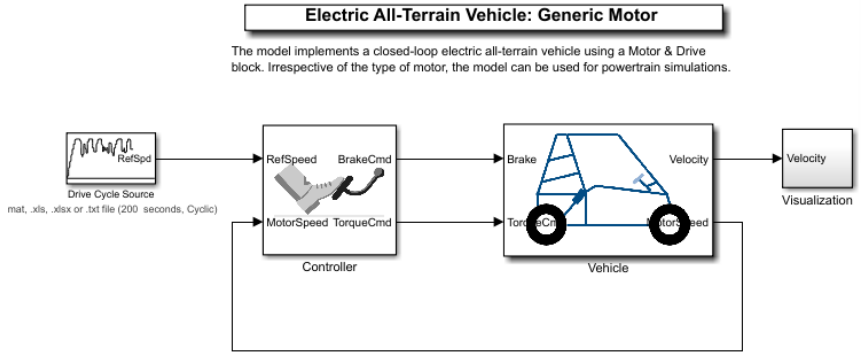
SOC estimation



Battery pack system design



Cell characterization

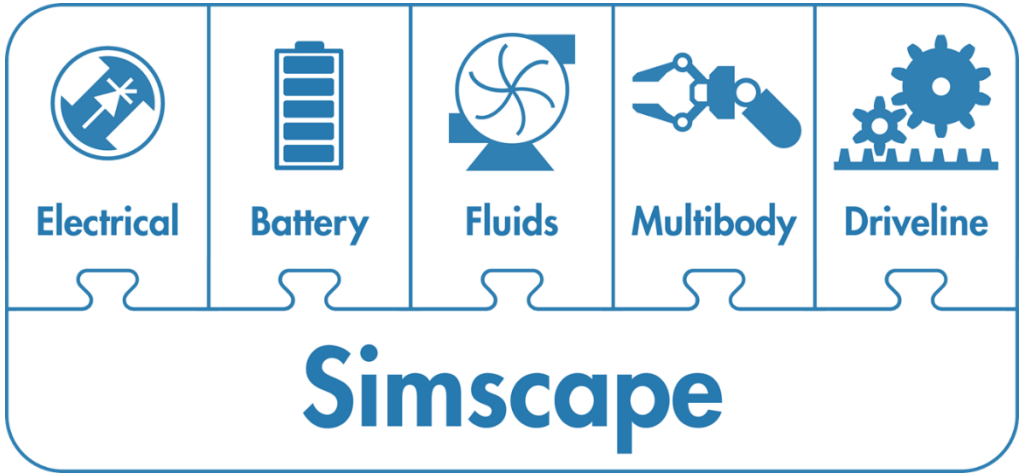


Copyright, 2022. The MathWorks, Inc.

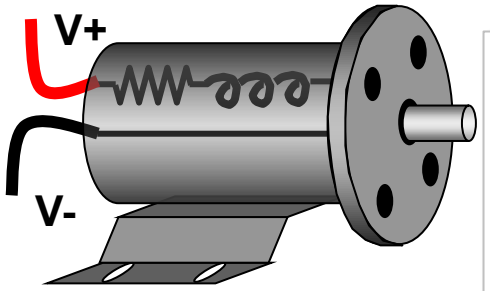
Agenda

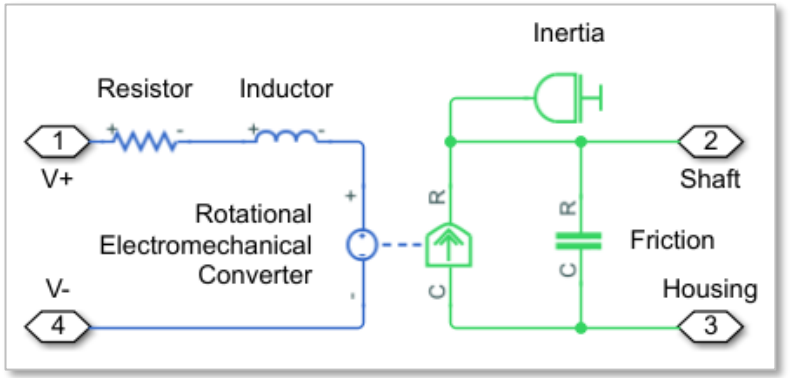
- Cell Characterization for BAJA Vehicle
- Battery Pack System Model
- State of Charge Estimation
- Electric Powertrain Design
- **BAJA SAEINDIA: MathWorks® Electric Powertrain Simulation Challenge 2026**

Simscape
Overview



- Enables physical modeling (acausal) of multidomain physical systems
  - Assemble a schematic
  - Equations derived automatically
  - Leverage MATLAB and Simulink

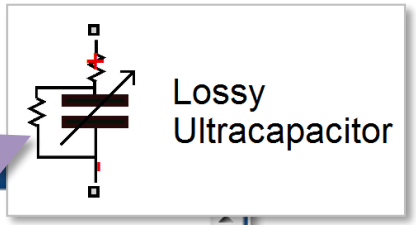




- With Simscape you can:
  - Refine requirements for system
  - Discover integration issues early
  - Design control systems and logic
  - Optimize system-level performance
  - Test embedded software without hardware prototypes

$$i = (C_0 + C_v v)\frac{dv}{dt} + \frac{v}{r_d}$$

```
Editor - C:\+MyComponent...ossyUltraCapacitor.ssc
40  equations
41    i == (C0 + Cv*vc)*vc.der + vc/Rd;
42    v == vc + i*R;
43  end
```



Understanding Physical Variables

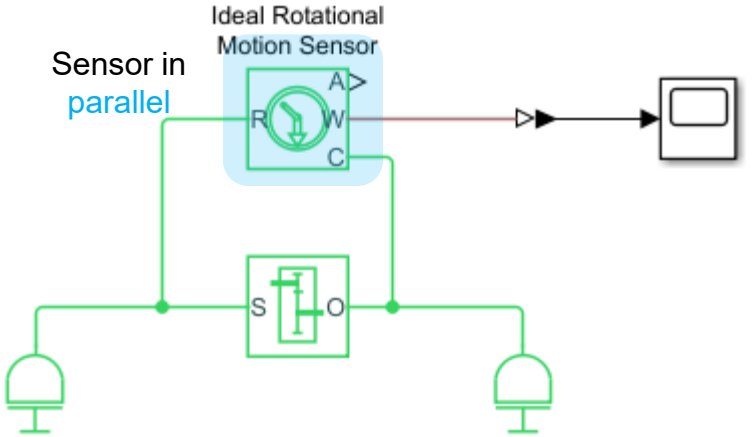
Energy flow = Across × Through

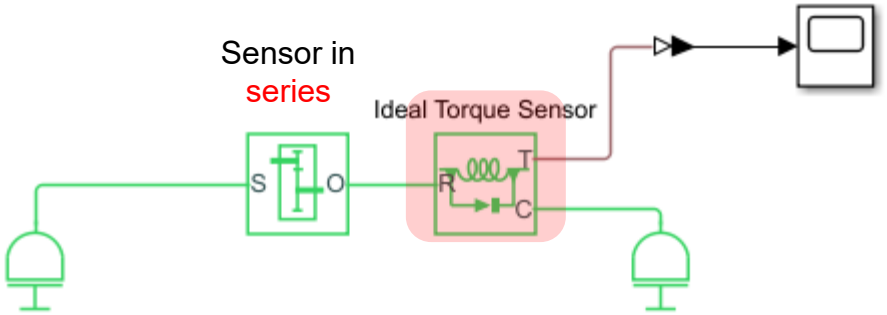
Sensor in parallel

Sensor in series

Power [W] = Voltage [V] × Current [A]

Power [W] = Angular Velocity [rad/s] × Torque [Nm]

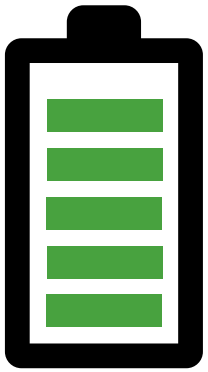


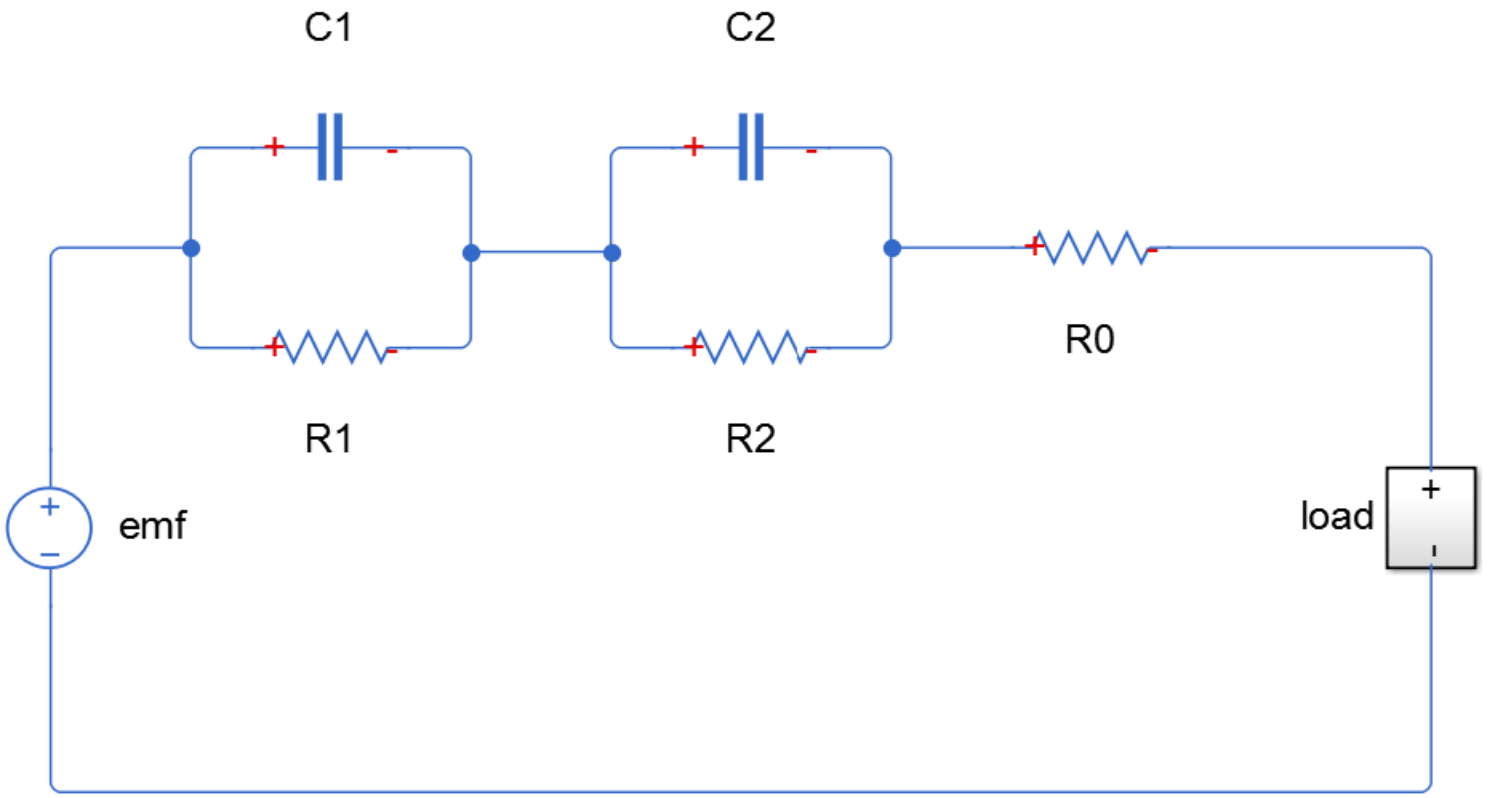


Across and Through Variables

Physical Domain	Across Variable	Through Variable
Electrical	Voltage	Current
Gas	Absolute pressure and temperature	Mass flow rate and energy flow rate
Hydraulic	Gauge pressure	Volumetric flow rate
Isothermal liquid	Absolute pressure	Mass flow rate
Magnetic	Magnetomotive force (mmf)	Flux
Mechanical rotational	Angular velocity	Torque
Mechanical translational	Translational velocity	Force
Moist Air	Absolute pressure, temperature, specific humidity (water vapor mass fraction), and trace gas mass fraction	Mixture mass flow rate, mixture energy flow rate, water vapor mass flow rate, and trace gas mass flow rate
Thermal	Temperature	Heat flow
Thermal liquid	Absolute pressure and temperature	Mass flow rate and energy flow rate
Two-phase fluid	Absolute pressure and specific internal energy	Mass flow rate and energy flow rate

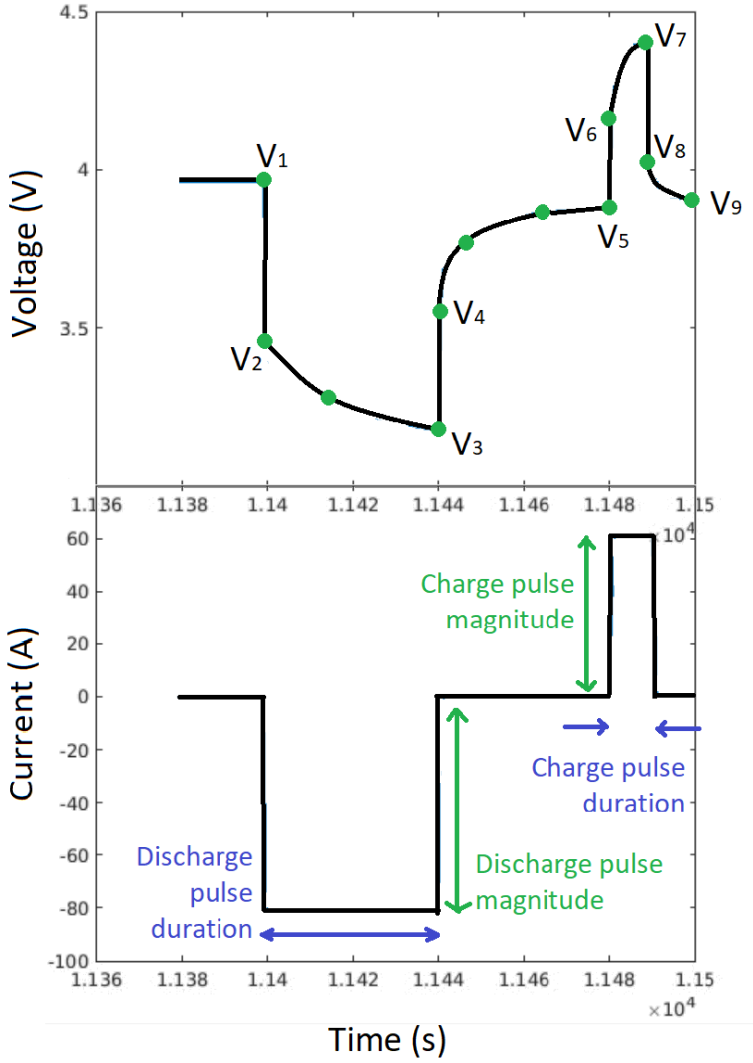
Equivalent Circuit of a Cell



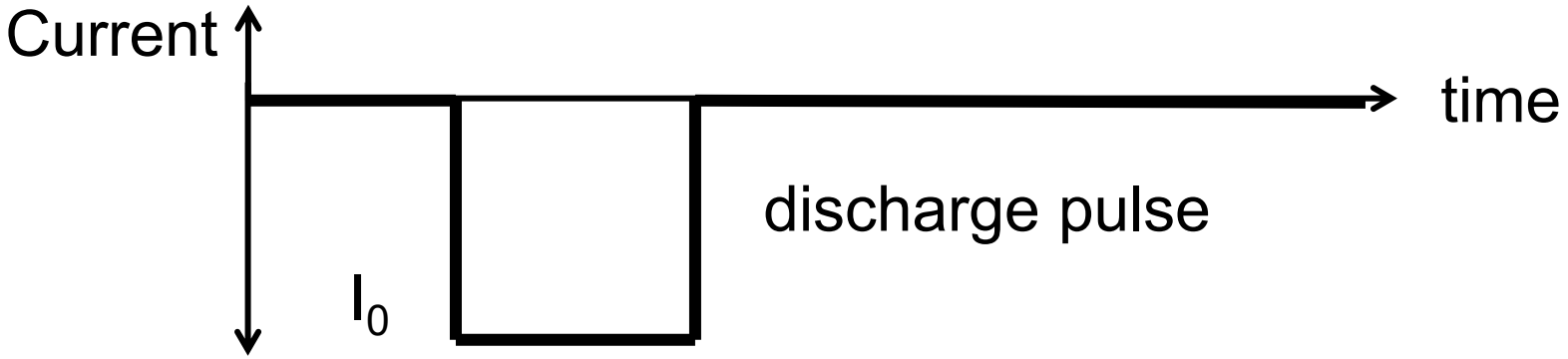


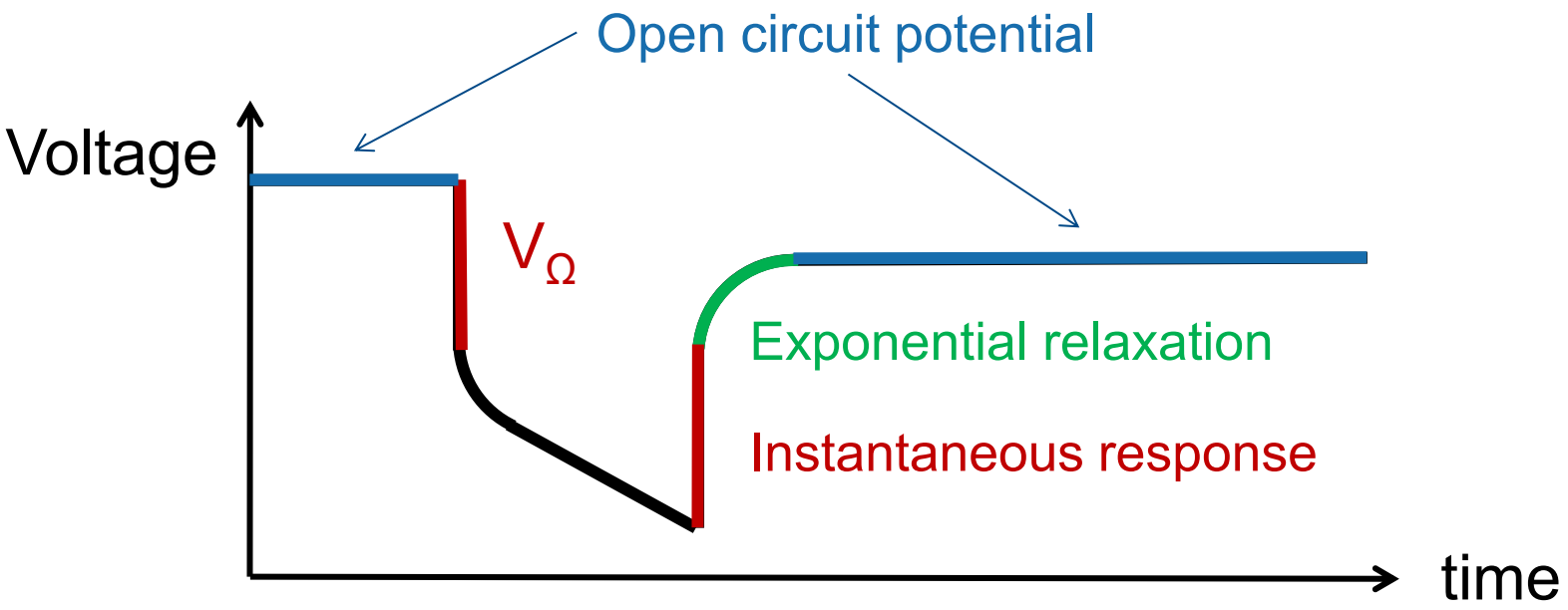
Hybrid Pulse Power Characterization (HPPC Test)

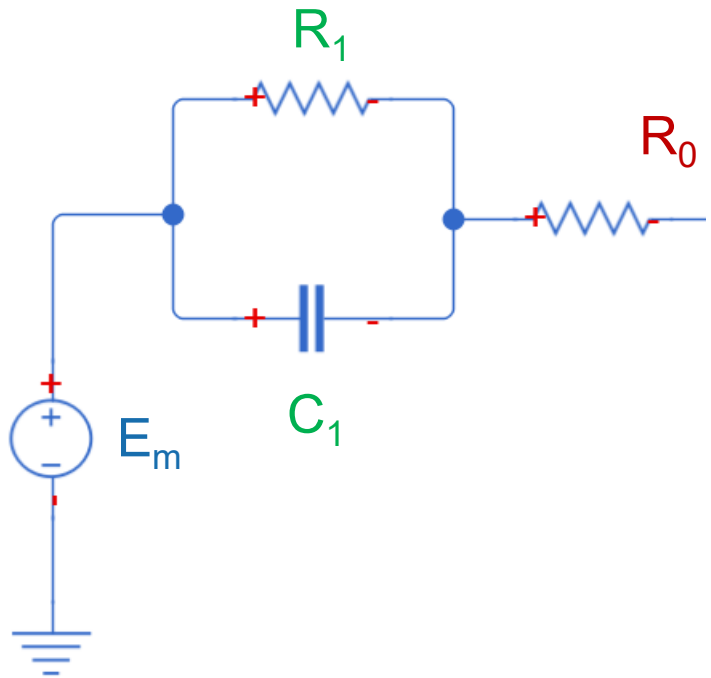
- A typical HPPC data is a set of discharge-charge pulses, applied to a battery at different state of charge (SOC) and at a given temperature.
- Typically, the test equipment is fully charged to undergo these pulse tests.
- At the end of every sequence, the SOC is discharged by a third of the C-rate.
- A long rest time of one hour is recommended for the cells to relax after every sequence of discharge-charge pulses.
- This process continues until it covers all points of interest in the SOC range.



Modeling Li-ion Battery Cells



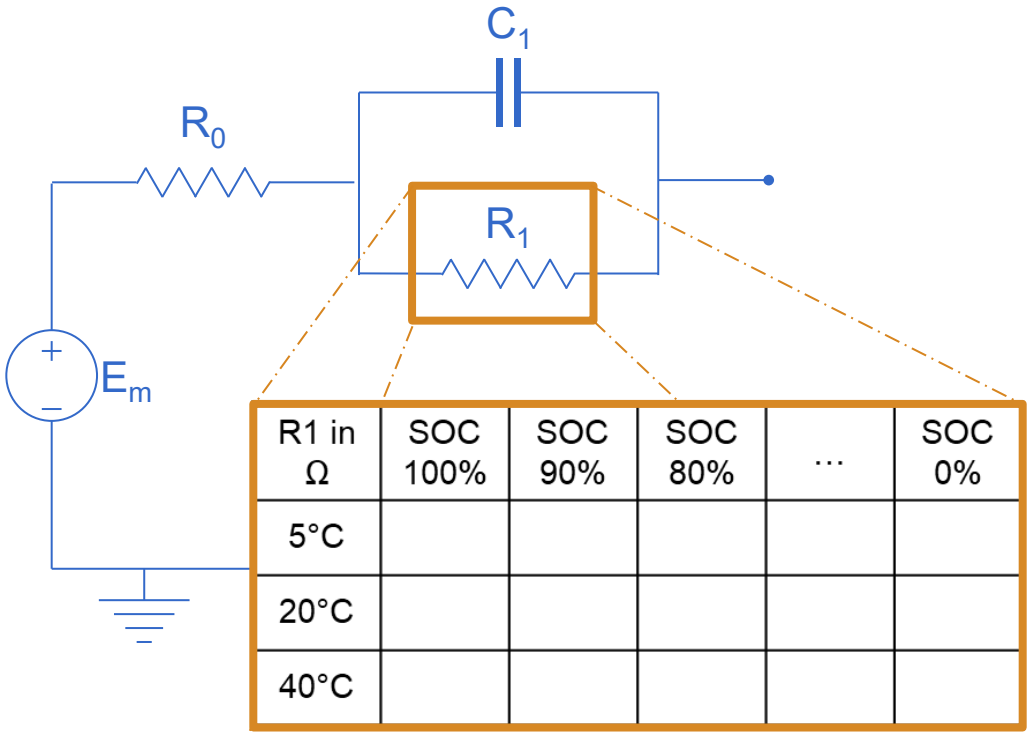


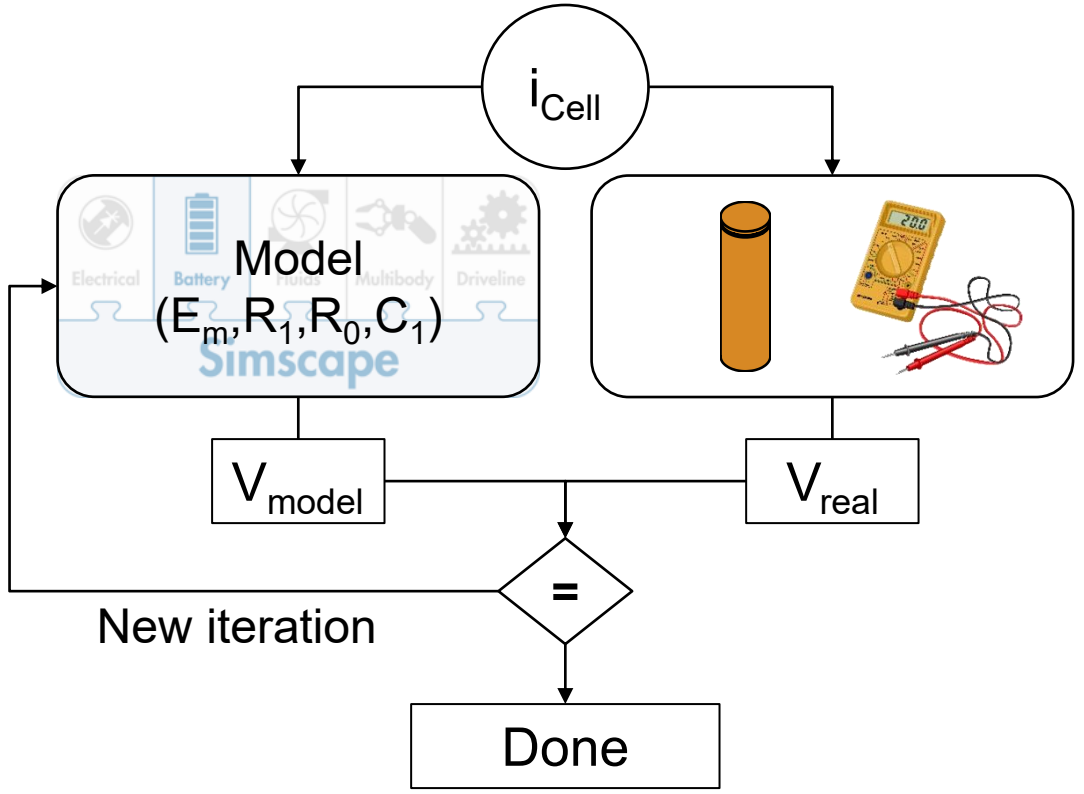


Parametrize the Cell Model

Curve Fitting Approach

- **Problem:** The cell parametrization requires a set of look-up tables
- **Solution:** The cell can be parametrized with measured data (using Curve Fitting)

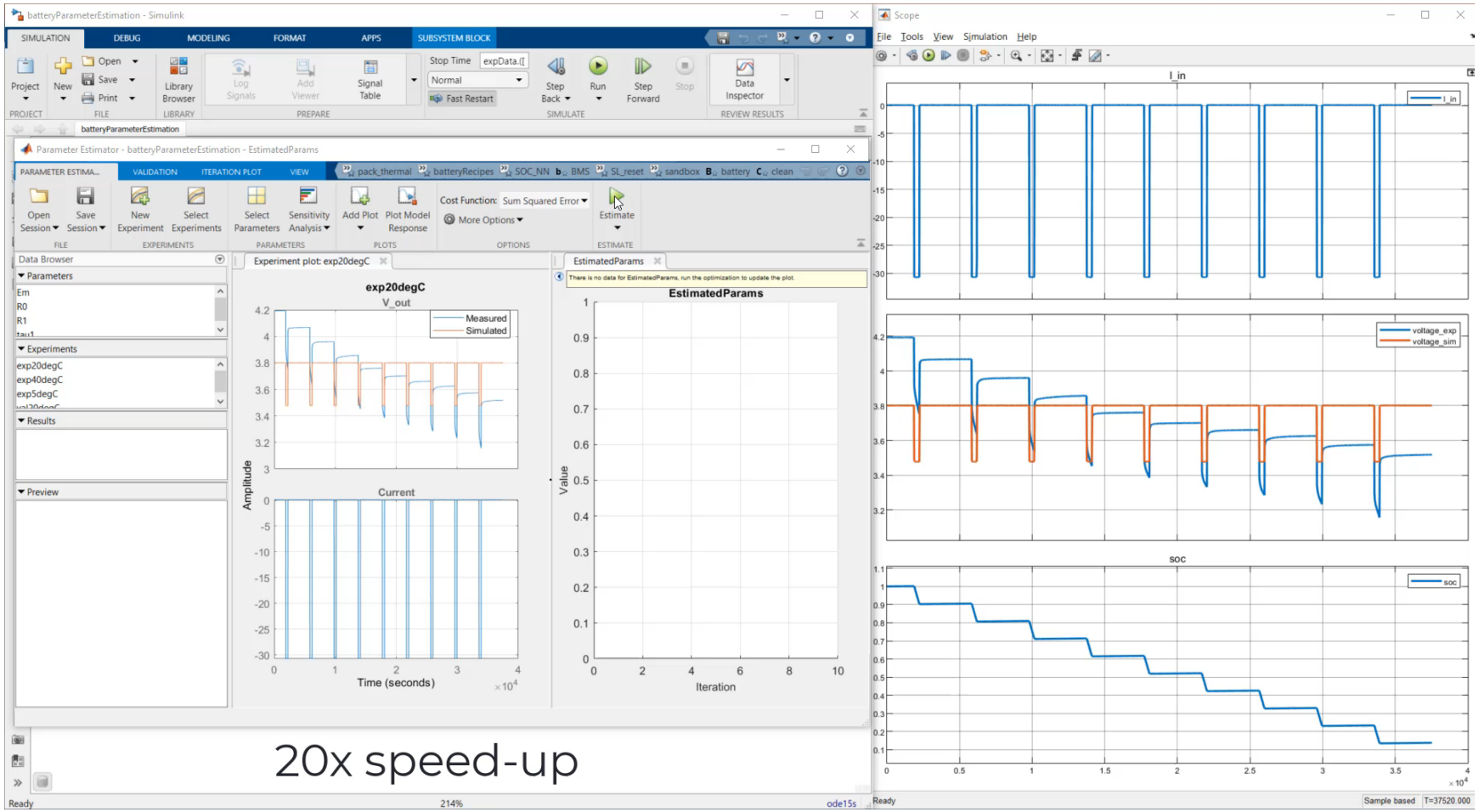




Demo: Cell Characterization

Parametrize the Cell Model

Circuit parametrization with Simulink Design Optimizer (SDO)



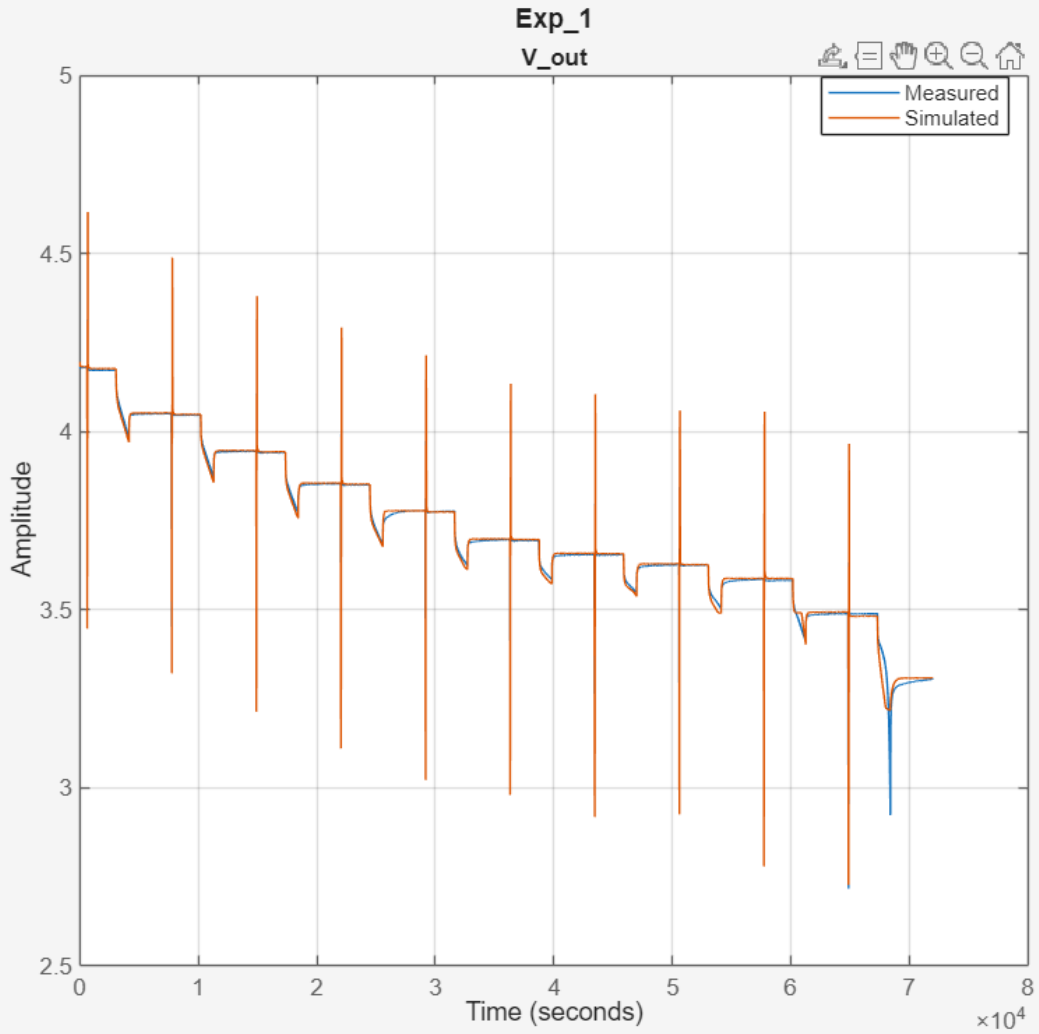
» Full video

Estimator* - batteryParameterEst2025b

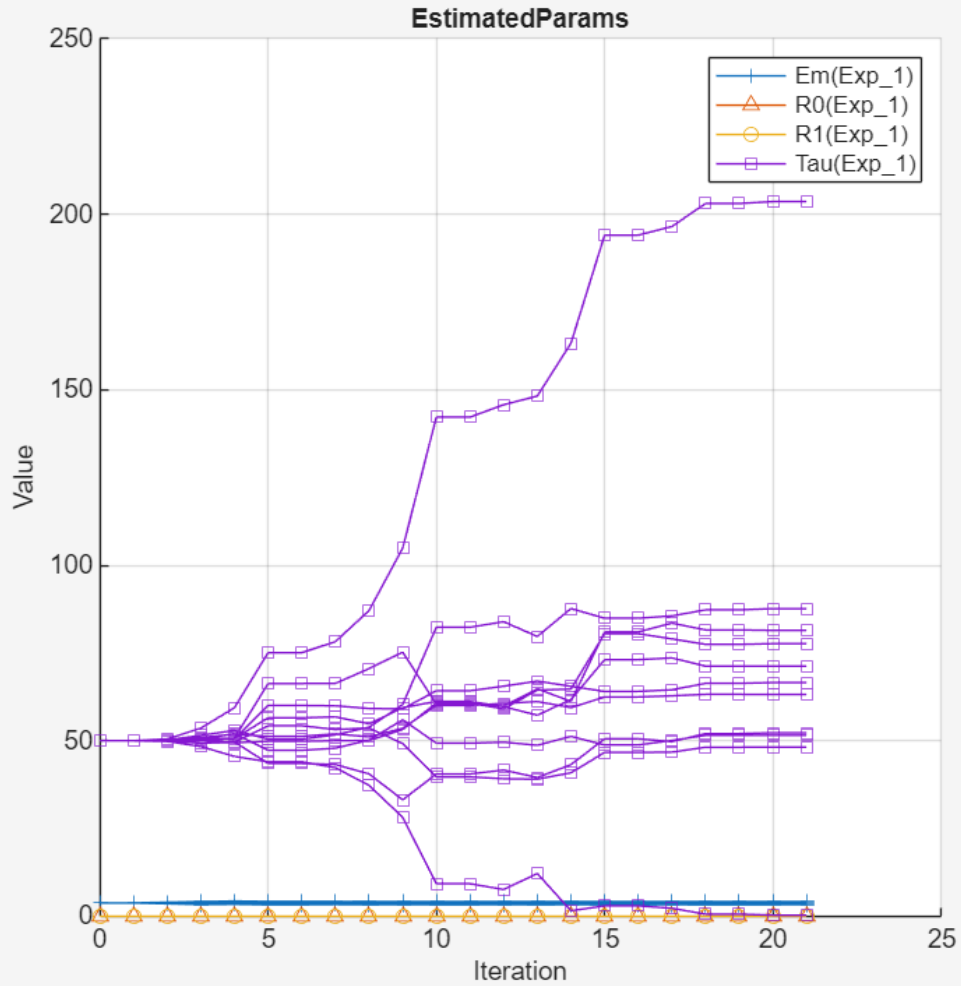
ESTIMATION | VALIDATION | EXPERIMENT PLOT

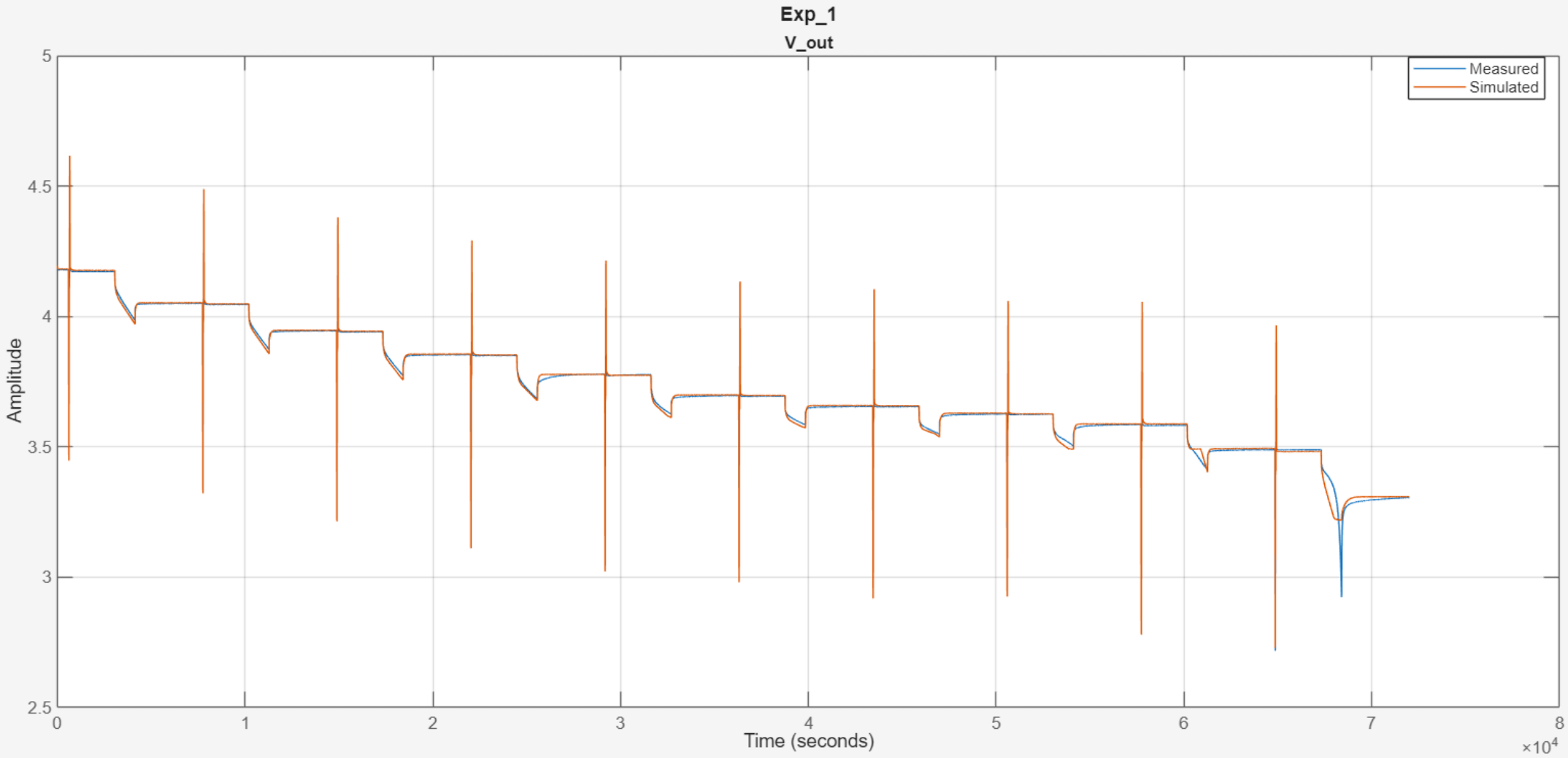
Save Session | New Experiment | Select Experiments (EXPERIMENTS) | Select Parameters | Sensitivity Analysis (PARAMETERS) | Add Plot | Plot Model Response (PLOTS) | Cost Function: Sum Squared Error | More Options (OPTIONS) | Estimate (ESTIMATE)

Exp_1



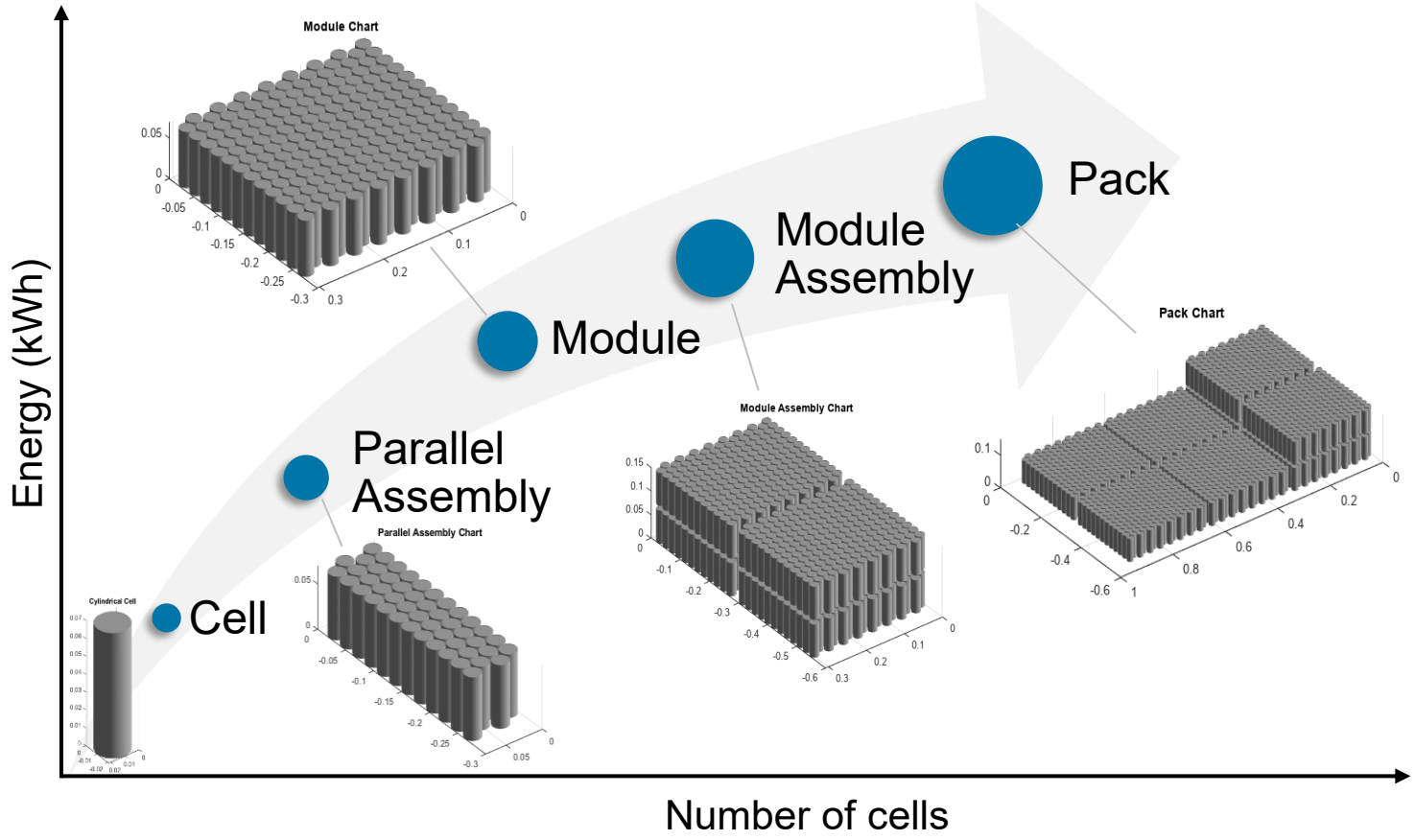
EstimatedParams



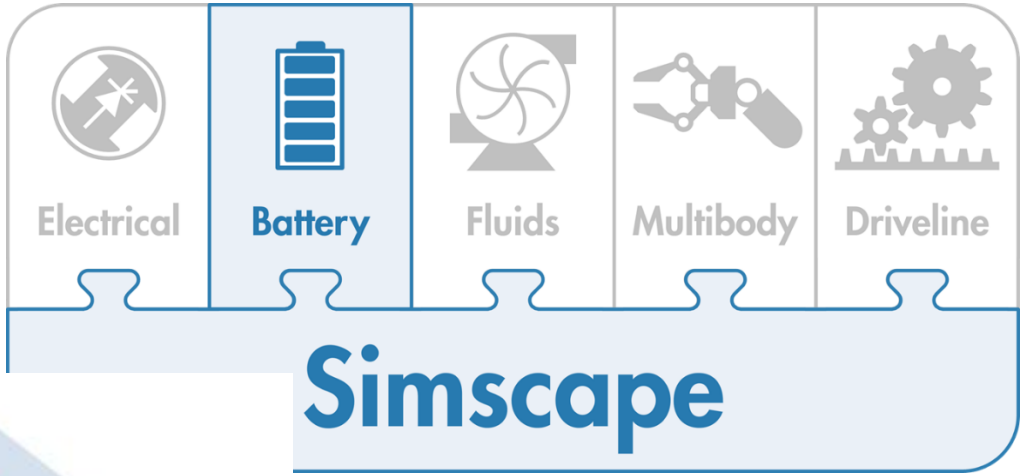

Exp_1
V_out
Measured
Simulated
Amplitude
Time (seconds)
×10^4
5
4.5
4
3.5
3
2.5
0
1
2
3
4
5
6
7
8

Battery Pack Builder

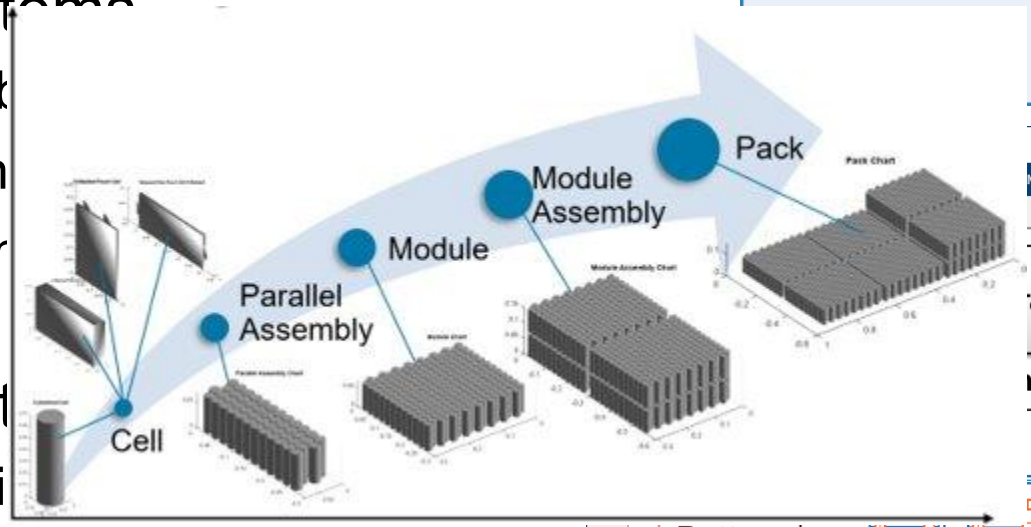
- Visualize the pack geometry and topology at each stage of the design.
- Define and visualize the simulation strategy.

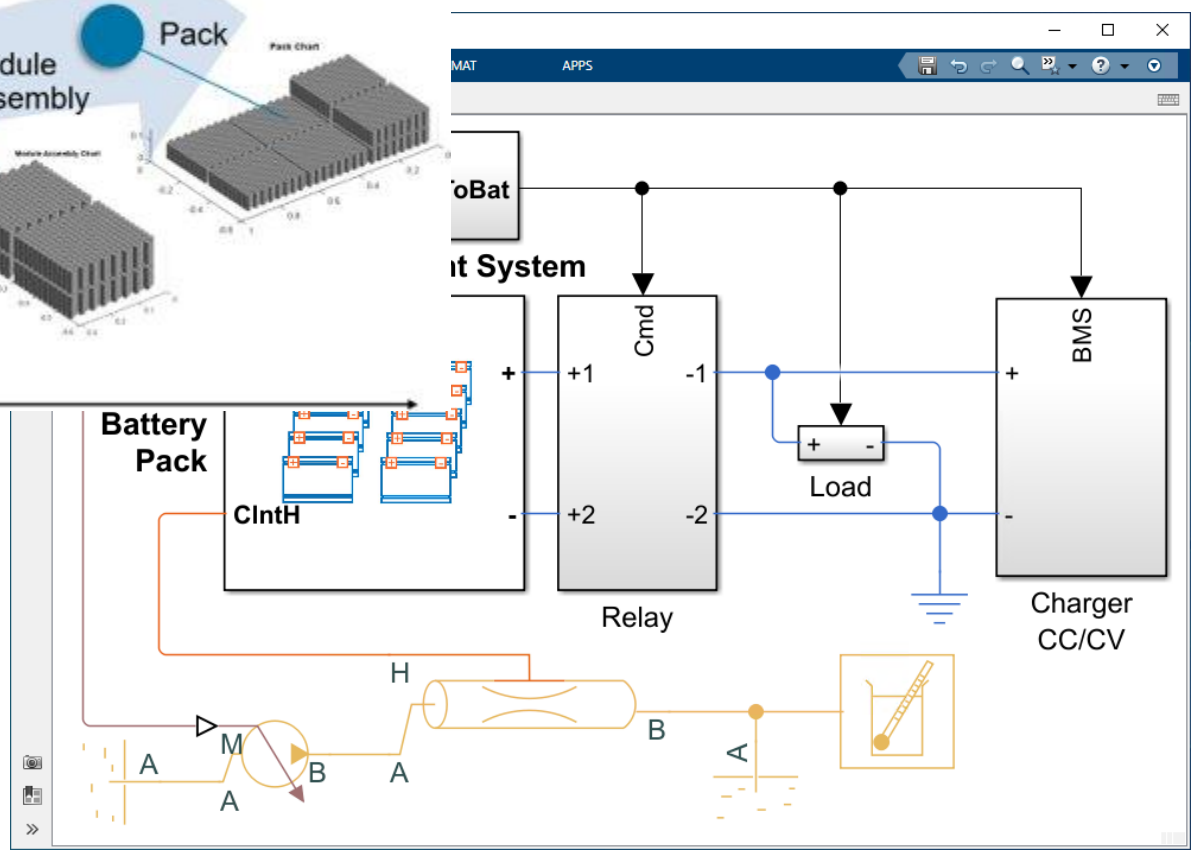


Simscape Battery



- Design and simulate battery and energy storage syst
  - Electrothermal cell b
  - Battery pack design
  - Battery managemer
- With Simscape Batt
  - Evaluate pack archi electrical and thermal requirements
  - Verify robustness of discharge, charge and thermal management algorithms
  - Validate algorithms using HIL testing

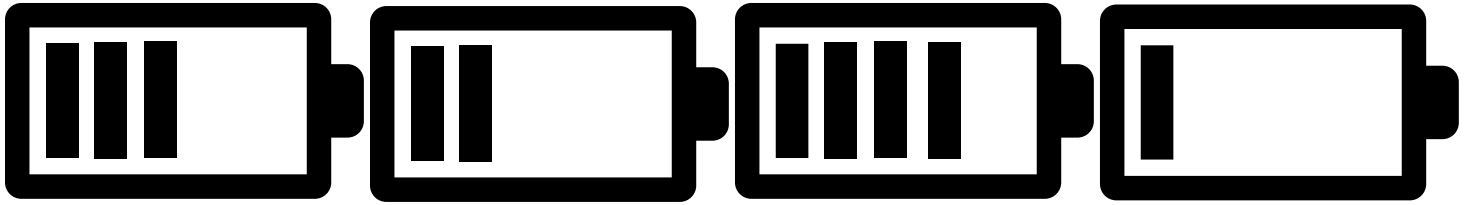




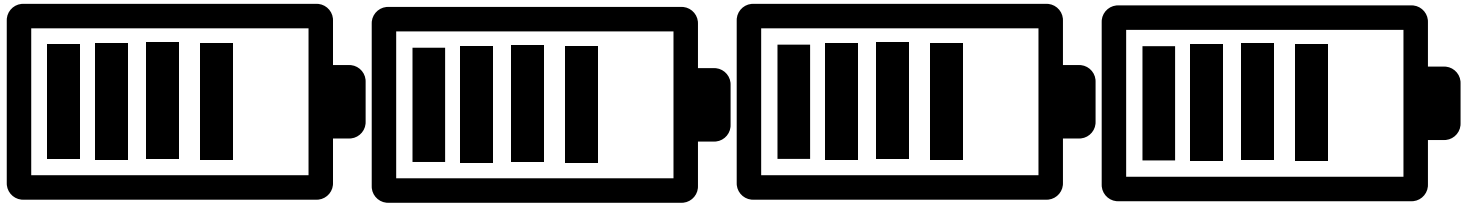
https://in.mathworks.com/products/simscape-battery.html

Hands-on Session 2: Battery Pack

Cell Balancing

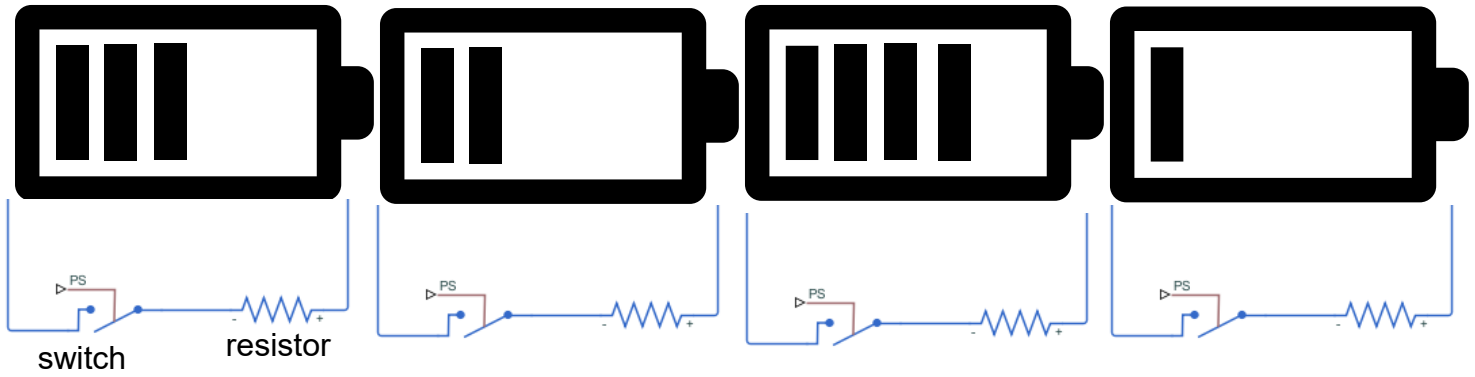


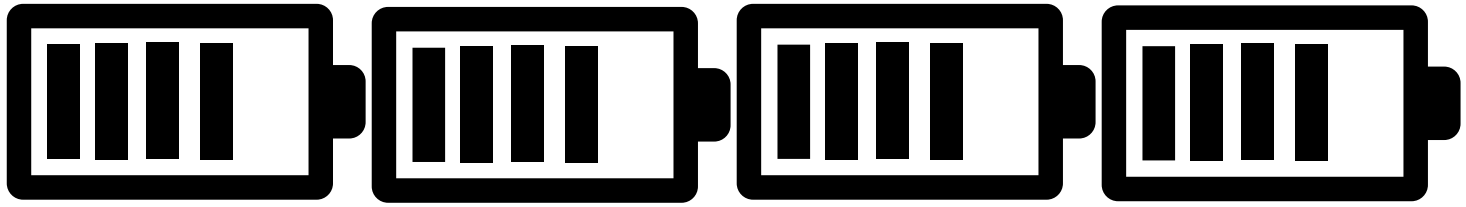
Undesirable situation



Desirable situation

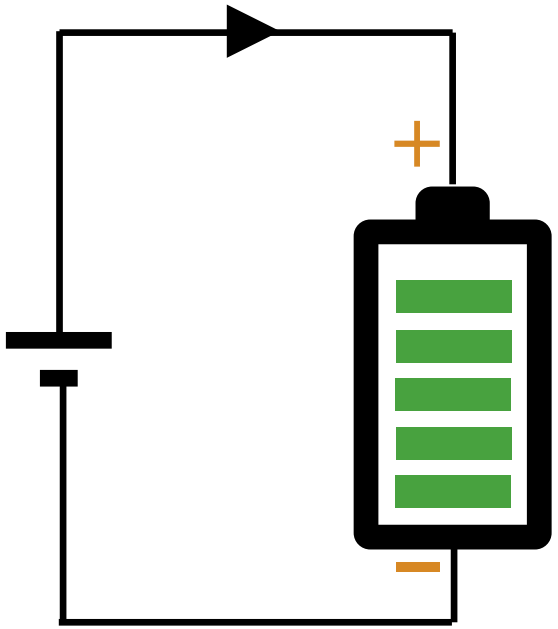
Passive Cell Balancing

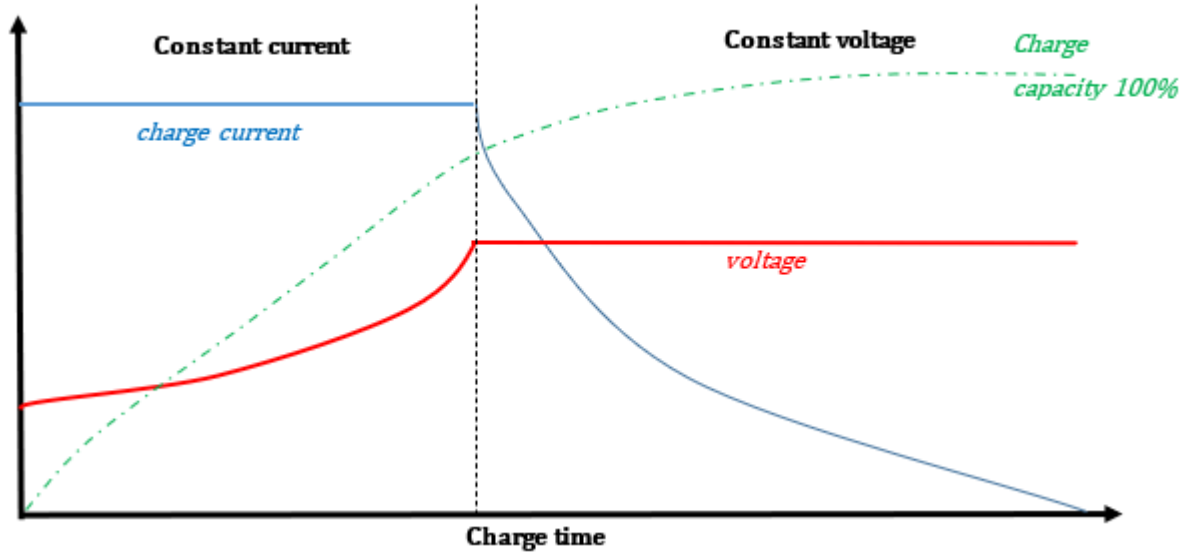




Balanced cell

CC CV Battery Charging



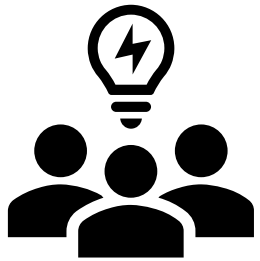


Hands-on Session 3: Passive Cell Balancing

State of Charge (SOC) Estimation

- Relative measure of the amount of energy stored in a battery, defined as the ratio between the amount of charge extractable from the cell at a specific point in time and the total capacity.
- Different methods
  - Open-Circuit Voltage (OCV)
  - Coulomb Counting
  - Kalman filtering
  - AI
  - Hybrid methods
- How can it help
  - BMS design
  - Develop control strategies
  - Improve battery life

- **What is the range of my vehicle?**
- **How many laps can I do?**
- **When do I need to swap the battery?**
- **How is the health of my battery?**

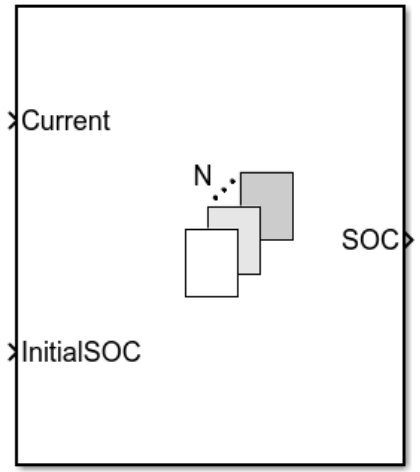




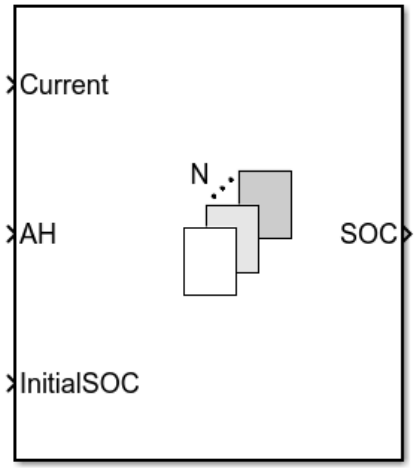
Credit: Team Stallions, Nirma University

State of Charge (SOC) Estimation

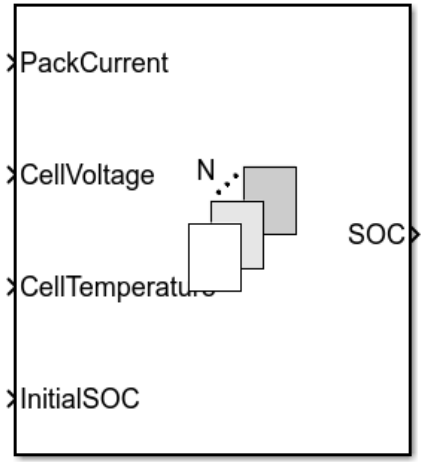
Simulink Blocks



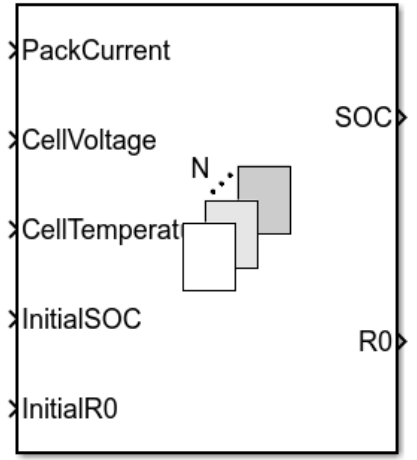
SOC Estimator
(Coulomb Counting)



SOC Estimator
(Coulomb Counting, Variable Capacity)



SOC Estimator
(Kalman Filter)



SOC Estimator
(Adaptive Kalman Filter)

Points to be considered in a racing competition

- Noisy race environment
- Decline in accuracy over a long endurance run
- Effect of temperature variation on battery
- Power consumption due to sudden change in acceleration and deceleration
- Battery degradation

**Kalman filter can address all the above. However, it is a complex algorithm and relies on accurate sensor measurement*

Demo: State of Charge

Demo: Electric Powertrain

BAJA SAEINDIA: MathWorks® Electric Powertrain Simulation Challenge 2026

Awards:

- 🥇 1st Place: ₹50,000
- 🥈 2nd Place: ₹40,000
- 🥉 3rd Place: ₹25,000

4th Place: ₹20,000

5th Place: ₹15,000

Stage 1 submission deadline
December 21, 2025



BAJA SAEINDIA: MathWorks® Electric Powertrain Simulation Challenge 2026

- Simulate and optimize your electric all-terrain vehicle's powertrain
- Work with a reference model and adapt it to your real vehicle specs
- Showcase your engineering skills and win exciting cash prizes!

Challenge Tasks:

- Design a detailed battery pack model
- Modify the powertrain subsystem to match your vehicle
- Demonstrate performance with simulation plots

Why Participate?

- Validate real-world performance
- Identify areas for fine tuning
- Back up your design with data
- Leave behind a legacy model
- Gain in-demand industry skills

Awards:

- 1st Place: ₹50,000
- 2nd Place: ₹40,000
- 3rd Place: ₹25,000

4th Place: ₹20,000
5th Place: ₹15,000

Key Dates

- Stage 1 submission deadline: **December 21, 2025**
- Stage 1 Results: December 27, 2025
- Stage 2 Q&A: During eBAJA physical event
- Final Results: At BAJA SAEINDIA 2026 award ceremony

For more details, follow this link: *to be provided by BAJA organizers*

© 2025 The MathWorks, Inc. MATLAB and Simulink are registered trademarks of The MathWorks, Inc. See *mathworks.com/trademarks* for a list of additional trademarks. Other product or brand names may be trademarks or registered trademarks of their respective holders.

Self-Paced Online Courses



Simscape Onramp

1 hour | Languages

Learn the basics of simulating physical systems in Simscape.



Multibody Simulation Onramp

12% | 2.5 hours | Languages

Learn the basics of modeling and simulating 3D mechanical systems using Simscape Multibody.



Power Systems Simulation Onramp

1.5 hours | Languages

Learn how to progressively build and validate power systems using Simscape Electrical.



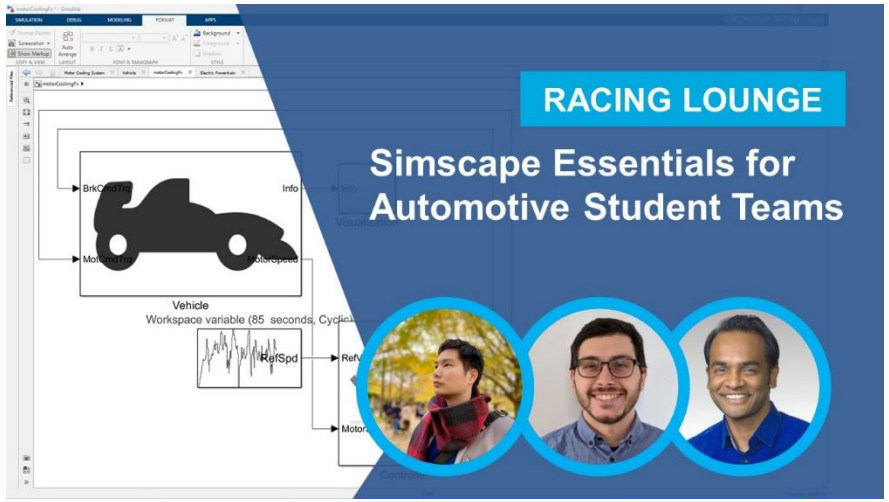
Battery Pack Modeling

1 hour | Languages

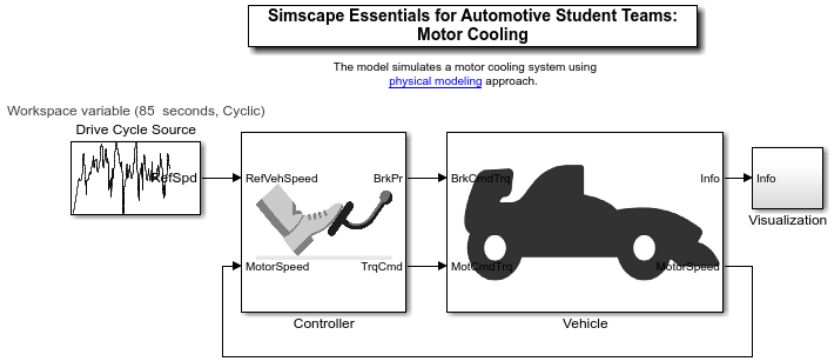
Use Simscape to implement cell-to-pack workflows, battery pack thermal modeling, and adding a cooling plate to battery packs.

Simscape Essentials for Automotive Student Teams

- Videos series for students to get started with the fundamentals of vehicle development for student competitions like Formula Student using Simscape™.
  - Longitudinal Vehicle Motion
  - Electric Powertrain
  - Motor Cooling System

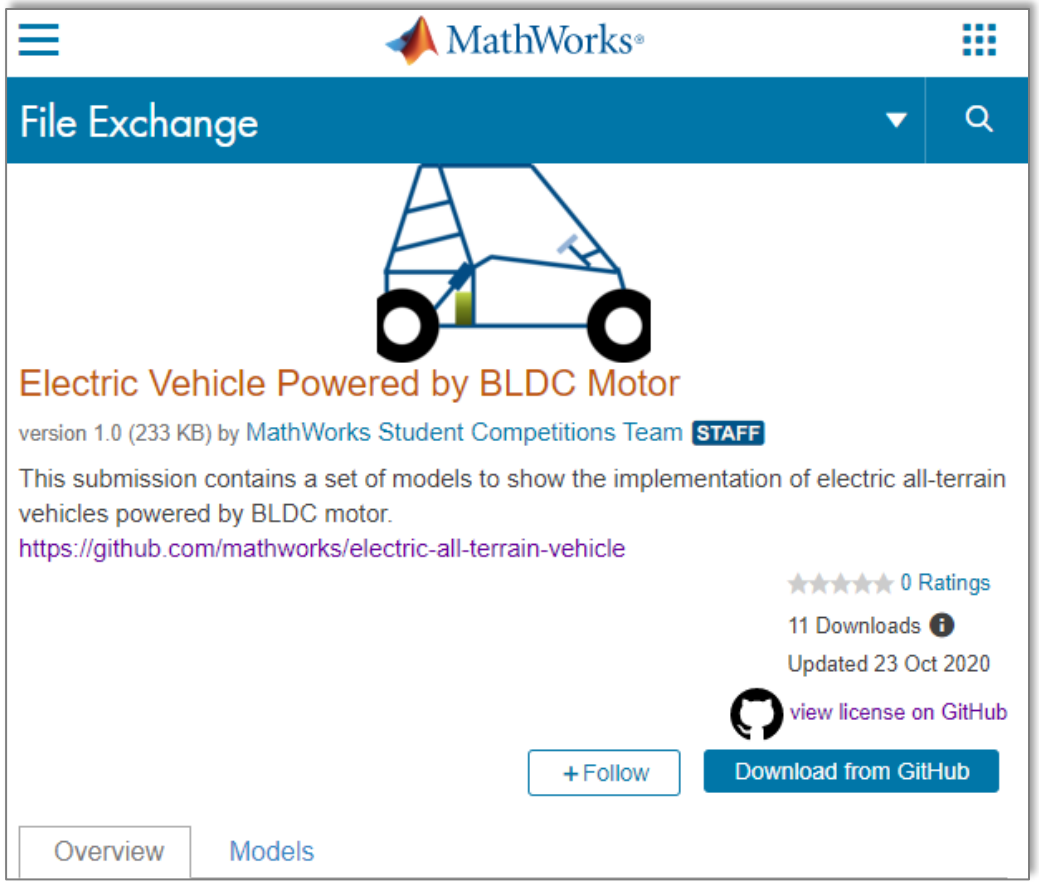


- Models are available on GitHub

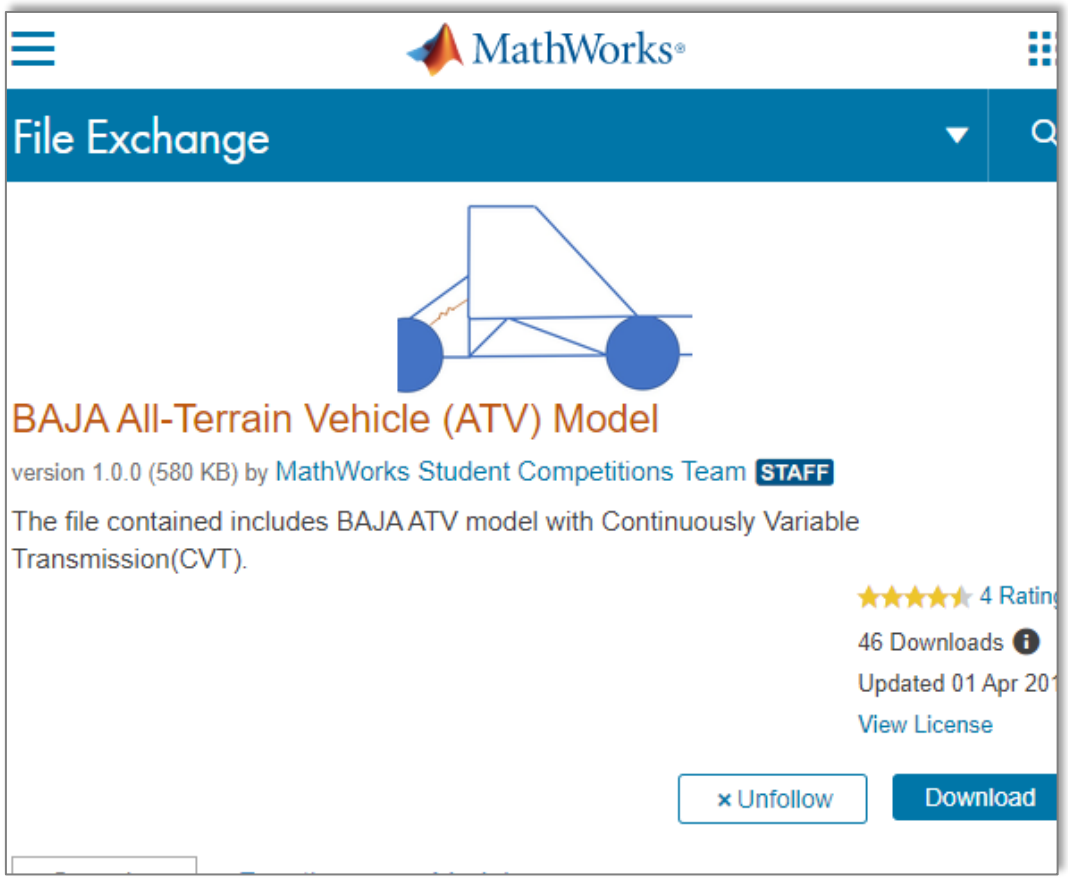


Copyright 2023 The MathWorks, Inc.

Download the Models: File Exchange and MathWorks GitHub



Link:
https://www.mathworks.com/matlabcentral/fileexchange/81623-electric-vehicle-powered-by-bldc-motor



Link:
https://www.mathworks.com/matlabcentral/fileexchange/70576-baja-all-terrain-vehicle-atv-model

Thank You

Veer Alakshendra



MATLAB and Simulink Racing Lounge



Contact us: racinglounge@mathworks.com